

Database Systems Lab Project Proposal

Restaurant POS System

1. Objective of the Proposal

Develop a Restaurant POS system to replace manual processes with a structured MySQL database, ensuring efficient, secure, and accurate operations.

2. Project Team

Abdul Moaiz Afridi

3. Introduction & Background

Restaurants require efficient order and sales management. Manual systems are slow and error-prone. A database-driven POS system improves speed, accuracy, and decision-making.

4. Problem Statement

- 1 Manual operations cause inefficiency
- 2 Errors in order and billing
- 3 Poor inventory tracking
- 4 No proper data analysis system

5. Proposed Solution

A web-based POS system using PHP and MySQL with real-time updates, secure login, and reporting features.

6. Objectives

- 1 Efficient data management
- 2 Fast retrieval
- 3 Data consistency
- 4 Secure access
- 5 Sales reporting

7. Scope

Includes order, user, inventory, and reporting modules. Excludes mobile apps and online payments.

8. Role of Database

- 1 Store structured data
- 2 Enable CRUD operations
- 3 Maintain integrity
- 4 Support reports

9. Features

- 1 Login system
- 2 Role-based access
- 3 Order management
- 4 Inventory control
- 5 Receipts
- 6 Reports

10. Data Design

Entities: Users, Products, Orders, OrderDetails

11. Existing Systems

Manual systems are inefficient. Proposed system is faster, reliable, and cost-effective.

12. Technology

Frontend: HTML/CSS Backend: PHP Database: MySQL Tools: XAMPP

13. Expected Outcomes

- 1 Improved efficiency
- 2 Reduced errors
- 3 Accurate billing
- 4 Better decisions